THE LOREM IPSUM THEOREM

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ABSTRACT. Lorem ipsum dolor sit amet: this paper is a placeholder, and it knows it. It exists to prove the road, not the theorem: the pipeline that turns a finished result into a folder holding a PDF, its LaTeX source, a sixty-second README, and a script that re-checks every number. A toy theorem rides along so that every part of the machine turns once: the level-L lorem carpet holds exactly 8^L cells, proved twice and machine-checked through level six.

1. Introduction

Take a square. Cut it into nine equal squares and throw away the middle one. Do the same to each of the eight survivors, and keep going. Figure 1 shows the second step: a square lace with 64 cells where 81 could stand. Mathematicians call the limit of this game the Sierpiński carpet; here, for one paper only, it is the *lorem carpet*.

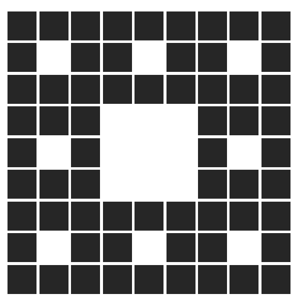

FIGURE 1. The level-two lorem carpet: $8^2 = 64$ cells where $9^2 = 81$ could stand.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua [2]. A real paper spends this paragraph telling a stranger why the question matters; this one is filler by design, and says so.

The question this paper pretends to ask: how many cells survive at each step? The answer is the cleanest kind, eight choices per step, multiplied. Theorem 3.1 states it, Section 4 proves it two ways, and Fact 3.2 reports the machine agreeing through level six.

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2. Definitions

Definition 2.1. For an integer $L \geq 0$ and $0 \leq x, y < 3^L$, write $x_i, y_i \in \{0, 1, 2\}$ for the i-th digits of x and y in base 3. The *level-L lorem carpet* is

$$C_L = \{(x, y) : \text{no position } i \text{ has } x_i = y_i = 1\}.$$

Remark 2.2. C_1 is the 3×3 grid with its center removed, since the center is the one cell with $x_0 = y_0 = 1$. Ut enim ad minim veniam, quis nostrud exercitation ullamco.

3. Results

Theorem 3.1. *For every $L \geq 0$, the level-L lorem carpet holds exactly 8^L cells: $|C_L| = 8^L$.*

Fact 3.2. For $0 \leq L \leq 6$, direct enumeration of every pair (x, y) with $0 \leq x, y < 3^L$ returns $|C_L| = 8^L$ exactly: 1, 8, 64, 512, 4096, 32768, 262144. Checked by `scripts/verify.py`.

Conjecture 3.3. *Heavier cargo will ride this road. Evidence: the road exists, and this paper travelled it end to end. Failure mode: the next paper never comes.*

4. Proofs

Proof of Theorem 3.1. A cell of C_L is exactly a choice, in each of the L digit positions, of a pair $(x_i, y_i) \in \{0, 1, 2\}^2 \setminus \{(1, 1)\}$. The positions are independent and each offers $9 - 1 = 8$ options, so $|C_L| = 8^L$.

Alternatively, by self-similarity: the leading digits (x_{L-1}, y_{L-1}) place a cell in one of the 8 surviving cells of C_1, and the remaining digits range over a full copy of C_{L-1} inside it. Hence $|C_L| = 8 \cdot |C_{L-1}|$ for $L \geq 1$, with $|C_0| = 1$. Two proofs, one count: the first explains the formula, the second explains the picture. □

5. Reproducibility

Two scripts, plain Python 3 with no dependencies. `scripts/verify.py` runs in a few seconds: it enumerates every cell of C_L for $0 \leq L \leq 6$, a 729×729 grid at the top, and asserts that the count equals 8^L, dying loudly on any mismatch. `scripts/figure.py` redraws Fig. 1 from the same rule. The sequence 8^n is OEIS A001018 [1].

Acknowledgments

This paper was developed and verified in collaboration with Claude (Anthropic). The author takes sole responsibility for every claim.

References

[1] OEIS Foundation Inc., Entry *A001018: Powers of 8*, The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A001018

[2] M. T. Cicero, *De finibus bonorum et malorum*, 45 BC. The quarry from which lorem ipsum was mined. https://www.thelatinlibrary.com/cicero/fin.shtml

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